Q1E | Train Polynomial Regression

Class: CS P556 - Applied Machine Learning
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Disclosure: This notebook was completed with the assistance of Claude Code (Opus 5.5) and the book *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd edition) by Aurélien Géron.

> E. Repeat everything from part D with polynomial regression and using SGD. Using validation loss, explore if your model overfits/underfits the data and report your findings. Ensure reproducibility by using a seed.

IMPORTANT

This notebook uses only the train dataset, so no modeling decision is influenced by the test data (data snooping). Every score below is a cross validation score on the training rows.

Answer Summary

Cross validation and metric

- Four-fold cross validation (`KFold` , shuffled with seed 42), scored by RMSE in `Life Ladder` points: it keeps the target's units and weighs large misses more than small ones. Validation rows are complete rows only, like the test file.
- Each choice starts from the simpler option, which another replaces only if its mean validation RMSE is at least 1% lower.

Features

- Same fixed shapes as Q1D, from Q1B on 7 attributes.

Polynomial regression: degree

- Every kept attribute's columns are expanded. By closed form (SVD), degree 2 is kept: 0.4980 against 0.5178 at degree 1, and degree 3 is 19.32% worse.

Missing values

- Dropping the 185 incomplete rows is kept: all 8 imputers are worse, the closest (KNN + missing flags) by 0.18%. Every model trains on the 1,374 complete rows.

Closed form vs SGD

- SGD at scikit-learn's defaults lands just above the closed form: 0.4987 against 0.4980.

Other hyperparameters

- `eta0` : only a narrow range works; 0.1 diverges, 0.001 is 2.08% worse, and the default 0.01 is kept.
- Schedule: the default `invscaling` , which shrinks the step from the first update, is kept; `constant` , `optimal` and `adaptive` all diverge.

Regularization

- A weak penalty changes nothing: at alpha 0.001 every penalty is within 0.0006 of no penalty.
- L1 bites first: at alpha 0.1 Lasso rises to 0.5380 while Ridge stays near no penalty.
- A strong penalty underfits: at alpha 10 Lasso and Elastic Net predict the mean.

Selected model

- Elastic Net at alpha 0.001 (`invscaling` , `eta0` 0.01, degree 2), picked as required: 0.4981 against 0.4987 without a penalty.
- Trained on the 1,374 complete rows and saved: training RMSE 0.4855.

Loss by epoch

- Most of the learning happens in the first 2 epochs; validation RMSE then edges down to 0.4981 and shows no lasting rise.
- The homework's loop restarts the step at 0.01 every epoch, so it flattens early and ends higher (0.5016).

Overfitting and underfitting

- Degree 1 underfits; from degree 3 the closed form overfits (training RMSE falls to 0.4083 while validation RMSE rises to 0.5942).
- The selected model fits well: a small gap between training and validation RMSE (0.4848 against 0.4981), no overfitting for a penalty to cure, and none from more epochs.

Reproducibility

- Seed 42 for the fold shuffle and every SGD model.

Imports

In [1]:
```
import joblib
import numpy as np
import pandas as pd
from IPython.display import display
from sklearn.compose import make_column_transformer
from sklearn.dummy import DummyRegressor
from sklearn.experimental import enable_iterative_imputer  # makes IterativeImputer
importable
from sklearn.impute import IterativeImputer, MissingIndicator, SimpleImputer
from sklearn.linear_model import SGDRegressor
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import KFold
from sklearn.pipeline import make_union

import Q1.consts as Q1_CONSTS
from Q1.utils import ScaledKNNImputer
from consts import SEED, CARRY_MARGIN, CV_SPLITS, MAX_ITER, TOL, POWER_T,
LEARNING_RATE_SCHEDULES, ETA0S, ALPHAS, L1_RATIO, DEGREES

from utils import (
        read_csv_to_df,
        STATISTICS,
        PLOTTING,
        PIPELINES,
)
```

```
pd.set_option('display.max_rows', None)
pd.set_option('display.min_rows', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', '{:.4f}'.format)  # 4 decimals, like
STATISTICS.results_df, in the tables it does not build
```

Fetch Data

In [2]:
```
train_df = read_csv_to_df(path=Q1_CONSTS.TRAIN_PATH)

ATTRIBUTES = [column for column in train_df.columns if column != Q1_CONSTS.TARGET]
X, y = train_df[ATTRIBUTES], train_df[Q1_CONSTS.TARGET]
```

Setup

Folds

In [3]:
```
# Validation rows are complete rows only, as in the test file. The incomplete rows are
added to the training side of
# every fold, so a model that imputes them and one that drops them are scored on the
same validation rows.
complete = np.flatnonzero(X.notna().all(axis=1))
incomplete = np.flatnonzero(X.isna().any(axis=1))
cuts = KFold(n_splits=CV_SPLITS, shuffle=True, random_state=SEED).split(complete)
FOLDS = [
        (np.concatenate([complete[train], incomplete]), complete[validation])
        for train, validation in cuts
]
DROP_FOLDS = [
        (train[np.isin(train, complete)], validation)
        for train, validation in FOLDS
]  # the same folds without the incomplete rows

pd.DataFrame({
        'folds': [len(FOLDS), len(DROP_FOLDS)],
        'training rows': [f'{min(len(t) for t, _ in folds)} to {max(len(t) for t, _ in
folds)}' for folds in (FOLDS, DROP_FOLDS)],
        'validation rows': [f'{min(len(v) for _, v in folds)} to {max(len(v) for _, v in
folds)}' for folds in (FOLDS, DROP_FOLDS)],
}, index=['FOLDS (incomplete rows imputed)', 'DROP_FOLDS (incomplete rows dropped)'])
```

Out[3]:

	folds	training rows	validation rows
FOLDS (incomplete rows imputed)	4	1215 to 1216	343 to 344
DROP_FOLDS (incomplete rows dropped)	4	1030 to 1031	343 to 344

Features

Attribute	Shape	Source
`Log GDP per capita`	raw	Q1B; raw chosen over its hinge during development, not shown
`Social support`	reciprocal, 1 / (1.2 - x) (`'reciprocal'`)	Q1B

Attribute	Shape	Source
`Healthy life expectancy at birth`	floor at 56.8 (`'floor'`)	Q1B; kept despite Q1C's r 0.851 with `Log GDP per capita` , chosen during development, not shown
`Freedom to make life choices`	raw	Q1B suggested a hinge at 0.8; raw chosen over it during development, not shown
`Generosity`	hinge at 0.305 (`'hinge'`)	Q1B's second shape; chosen over Q1B's flag during development, not shown
`Perceptions of corruption`	raw	Q1B found no shape
`Positive affect`	raw	Q1B found no shape
`Negative affect`	dropped	chosen during development, not shown

In [4]:
```
GDP, SOCIAL, LIFE_EXP, FREEDOM, GENEROSITY, CORRUPTION, POS_AFFECT, NEG_AFFECT =
ATTRIBUTES
FITS = {attribute: STATISTICS.shape_fits(df=train_df, feature=attribute,
target=Q1_CONSTS.TARGET) for attribute in ATTRIBUTES}

def shape(*, attribute, name):
        """Q1B's shape of an attribute by name (ex 'hinge'), with the setting Q1B found,
as a pipeline step."""
        return next(fit for fit in FITS[attribute] if fit.name == name).transformer

SHAPES = {
        GDP: 'passthrough',  # Q1B; raw chosen over its hinge during development, not
shown
        SOCIAL: shape(attribute=SOCIAL, name='reciprocal'),  # Q1B
        LIFE_EXP: shape(attribute=LIFE_EXP, name='floor'),  # Q1B; kept despite r 0.851
with GDP (Q1C), chosen during development, not shown
        FREEDOM: 'passthrough',  # Q1B suggested a hinge at 0.8; raw chosen over it
during development, not shown
        GENEROSITY: shape(attribute=GENEROSITY, name='hinge'),  # Q1B's second shape;
chosen over Q1B's flag during development, not shown
        CORRUPTION: 'passthrough',  # Q1B found no shape
        POS_AFFECT: 'passthrough',  # Q1B found no shape
        NEG_AFFECT: 'drop',  # chosen during development, not shown
}
```

Runs

In [5]:
```
RUNS = {}  # every cross validation result, keyed by name

def build(*, model, imputer=None, degree=1):
        """The pipeline of one experiment on the features: imputer (none by default: the
incomplete rows are dropped),
        shapes, standardized columns, polynomial terms above degree 1, model (see
PIPELINES.build_pipeline)."""
        return PIPELINES.build_pipeline(model=model, attributes=ATTRIBUTES,
shapes=SHAPES, imputer=imputer, degree=degree)

def score(*, name, pipeline, folds=DROP_FOLDS):
        """Cross validate a pipeline and keep its result under name. The default folds
leave the incomplete rows out, to go
        with build's default."""
        RUNS[name] = PIPELINES.cv_rmse(pipeline=pipeline, X=X, y=y, folds=folds)

def table(*, names, baseline=None):
        """One row per run in names: mean validation RMSE, mean training RMSE, the most
epochs for SGD, and the change from
        the baseline when one is given. A run more than ten times Dummy's error shows as
diverged (Method)."""
```

```
        return STATISTICS.results_df(
                runs={name: RUNS[name] for name in names},
                baseline=baseline,
                diverged_above=10 * RUNS['Dummy']['validation rmse'].mean(),
        )

def lowest(*, names):
        """The run in names with the lowest mean validation RMSE."""
        return min(names, key=lambda name: RUNS[name]['validation rmse'].mean())

def carried(*, baseline, names):
        """The decision rule: the run in names with the lowest mean validation RMSE
replaces the baseline (the simpler
        option) only if it is at least CARRY_MARGIN (1%) lower; otherwise the baseline
is kept."""
        best = lowest(names=names)
        gain = 1 - RUNS[best]['validation rmse'].mean() / RUNS[baseline]['validation
rmse'].mean()
        return best if gain >= CARRY_MARGIN else baseline
```

Method

- **Metric:** RMSE in `Life Ladder` points, because it is in the target's own units and weighs large misses more than small ones, and it is the score F reports on the test set. `validation rmse` is the mean over the 4 folds of the RMSE on each fold's validation rows, rows the model never saw; every choice is made on it. `train rmse` is the same on the rows each model was fit on: far below `validation rmse` means the model fits its own rows better than new ones. A run whose mean validation RMSE is more than ten times Dummy's (predicting the training mean) is shown as `diverged`.
- **Folds:** four-fold cross validation (`KFold`), shuffled with seed 42. Validation rows are the 1,374 complete rows only, like the test file, which Q1A found has no missing values. The 185 incomplete rows sit on the training side when an imputer fills them and are left out when they are dropped, so every option is scored on the same validation rows.
- **Rule:** each choice starts from a baseline, the simpler option: degree 2, dropping the incomplete rows, and scikit-learn's default `eta0` (0.01) and schedule (`invscaling`). The option with the lowest mean validation RMSE replaces the baseline only if it is at least 1% lower (`CARRY_MARGIN`); a smaller gain is not worth the added step. The `% vs baseline` column shows the change. The selected regularized model has no baseline: the lowest mean wins. Degree 1 is D's model, not polynomial regression: it is shown for reference but cannot be carried.
- **Pipeline:** imputer (only when one is carried), shapes, every attribute's columns standardized (the missing flags stay 0/1), polynomial terms, model. The target is standardized too, so `alpha` and `tol` mean the same thing in Q1 and Q2 (Q1D's Why the Target Is Standardized shows what this changes), and the predictions come back in `Life Ladder` points. Every step is fit on the training rows of a fold only.
- **Polynomial terms:** at degree d, the standardized columns are expanded into every product of up to d of them (`PolynomialFeatures` with `include_bias=False`: no column of 1s, since the model fits its own intercept), then standardized again. Every kept attribute's columns are expanded; the missing flags are not.
- **Closed form:** least squares solved by SVD (`PIPELINES.SVDRegression`, numpy's `lstsq`). Some expanded columns are exact combinations of others: `LinearRegression` returns huge coefficients on them, while SVD returns the solution with the smallest coefficients.
- **Seed:** seed 42 for every step that has randomness: the fold shuffle, the iterative imputer and every SGD model. Dummy, SVD, mean, median and KNN are deterministic.

- **Optimism:** every choice is made on these folds, and Q1B's shape settings were found on all training rows, so the scores are a little optimistic. The test set gives the honest estimate.

Baselines

Dummy predicts the training mean of each fold, with the incomplete rows dropped (the baseline for missing values).

In [6]:
```
score(name='Dummy', pipeline=build(model=DummyRegressor()))
YMAX = 1.1 * RUNS['Dummy']['validation rmse'].mean()  # a run that diverges is drawn at the top of a chart instead of flattening the rest

table(names=['Dummy'])
```

Out[6]:

	validation rmse	train rmse
Dummy	1.1281	1.1279

Notes / Observations

- **The reference every degree is read against:** predicting the training mean scores 1.1281.

Degree

The closed form on the features at degrees 1 to 4, with the incomplete rows dropped. Degree 2 is the baseline (rule in Method); degree 1 is D's model, shown for reference only. Training and validation RMSE side by side show underfitting (both high and close) and overfitting (training RMSE falling while validation RMSE rises).

In [7]:
```
DEGREE_RUNS = {degree: f'Closed form, degree {degree}' for degree in DEGREES}
for degree, name in DEGREE_RUNS.items():
        score(name=name, pipeline=build(model=PIPELINES.SVDRegression(), degree=degree))

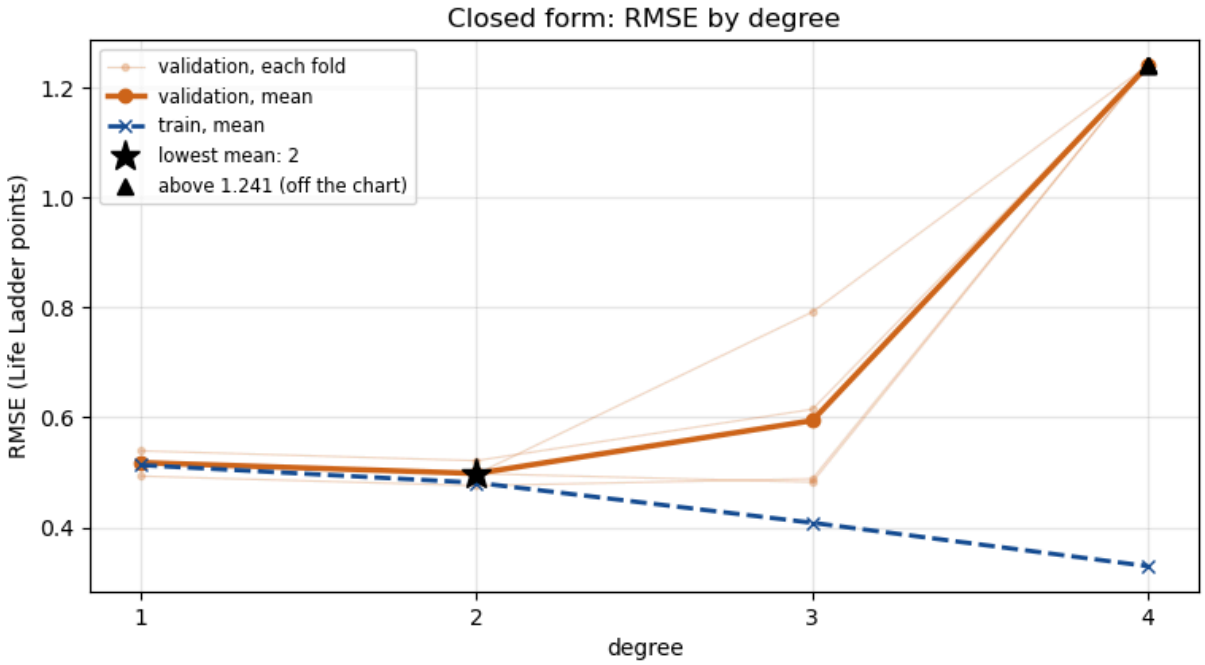
display(table(names=['Dummy', *DEGREE_RUNS.values()], baseline=DEGREE_RUNS[2]))
PLOTTING.parameter_search(
        runs={degree: RUNS[name] for degree, name in DEGREE_RUNS.items()},
        parameter='degree',
        ymax=YMAX,
        title='Closed form: RMSE by degree',
        unit=Q1_CONSTS.TARGET_UNIT,
);

POLYNOMIAL = [DEGREE_RUNS[degree] for degree in DEGREES if degree > 1]  # degree 1 is not polynomial regression
DEGREE = next(degree for degree, name in DEGREE_RUNS.items() if name ==
carried(baseline=DEGREE_RUNS[2], names=POLYNOMIAL))
pd.Series({'degree': DEGREE}, name='carried').to_frame()
```

	validation rmse	train rmse	% vs baseline
Dummy	1.1281	1.1279	+126.52%
Closed form, degree 1	0.5178	0.5132	+3.97%
Closed form, degree 2	0.4980	0.4811	baseline
Closed form, degree 3	0.5942	0.4083	+19.32%
Closed form, degree 4	3.3463	0.3293	+571.92%

Out[7]:

	carried
degree	2



Notes / Observations

- **Degree 2 is kept:** it is the baseline, and degrees 3 and 4 are both worse (degree 3: +19.32%).
- **Degree 1 underfits:** degree 2's squares and products lower training and validation RMSE together (validation: 0.5178 to 0.4980).
- **From degree 3 the closed form overfits:** training RMSE keeps falling (0.4083 at degree 3) while validation RMSE rises, up to 3.3463 at degree 4.

Missing Values

Ways to handle the 185 incomplete training rows, all by closed form on the features at the carried degree:

- **Drop incomplete rows (baseline):** train on the 1,374 complete rows only. The simpler option: no imputer. It is the carried degree's closed form run of Degree (`Closed form, degree 2`), so the table shows it under that name.
- **Mean:** fill with the attribute's training mean.
- **Median:** fill with the attribute's training median, which skewed values pull less than the mean.
- **KNN:** fill with the mean of the 5 most similar rows, measured on standardized attributes and returned in the attributes' own units (`ScaledKNNImputer` ; the shapes need those units).

- **Iterative:** predict each missing value from the other attributes, in rounds (`IterativeImputer`).
- **+ missing flags:** each fill again, plus a 0/1 column per kept attribute with missing values, marking the filled rows. The flags enter the model as 0/1: not standardized or expanded.

The imputer with the lowest mean validation RMSE replaces dropping only if it is at least 1% lower (Method). If dropping is kept, every later model trains on the complete rows only.

In [8]:
```
KEPT = [attribute for attribute in ATTRIBUTES if SHAPES[attribute] != 'drop']  # the
attributes the model keeps

def with_flags(*, imputer):
        """The imputer, plus a 0/1 column per kept attribute with missing values,
marking the filled rows."""
        return make_union(imputer, make_column_transformer((MissingIndicator(), KEPT)))
# the filled attributes come first, so the shapes find them by position

IMPUTERS = {DEGREE_RUNS[DEGREE]: None}  # dropping the incomplete rows is the carried
degree's closed form run of Degree: no imputer, already scored
for name, imputer in {
        'Mean': SimpleImputer(strategy='mean'),
        'Median': SimpleImputer(strategy='median'),
        'KNN': ScaledKNNImputer(),
        'Iterative': IterativeImputer(random_state=SEED),
}.items():
        IMPUTERS[name] = imputer
        IMPUTERS[f'{name} + missing flags'] = with_flags(imputer=imputer)
for name, imputer in IMPUTERS.items():
        if imputer is not None:
                score(name=name, pipeline=build(model=PIPELINES.SVDRegression(),
imputer=imputer, degree=DEGREE), folds=FOLDS)

display(table(names=list(IMPUTERS), baseline=DEGREE_RUNS[DEGREE]))
CLOSED_FORM = carried(baseline=DEGREE_RUNS[DEGREE], names=IMPUTERS)  # the carried
closed form, the one SGD is compared against
IMPUTER = IMPUTERS[CLOSED_FORM]
CV_FOLDS = DROP_FOLDS if IMPUTER is None else FOLDS
print(f'carried: {CLOSED_FORM}, incomplete rows dropped' if IMPUTER is None else
f'carried: {CLOSED_FORM}')
```

	validation rmse	train rmse	% vs baseline
Closed form, degree 2	0.4980	0.4811	baseline
Mean	0.5020	0.4902	+0.79%
Mean + missing flags	0.5010	0.4856	+0.61%
Median	0.5027	0.4904	+0.93%
Median + missing flags	0.5013	0.4858	+0.66%
KNN	0.4993	0.4909	+0.25%
KNN + missing flags	0.4989	0.4871	+0.18%
Iterative	0.5000	0.4941	+0.41%
Iterative + missing flags	0.4994	0.4901	+0.28%

```
carried: Closed form, degree 2, incomplete rows dropped
```

Notes / Observations

- **Dropping the incomplete rows is kept, and every imputer is worse:** the closest, KNN + missing flags, is 0.18% higher, so every later model trains on the 1,374 complete rows.

- **The simple fills cost the most:** mean and median are the worst (Median: +0.93%); KNN and iterative, which fill from the other attributes, come closer.
- **Missing flags help every fill a little** (Mean: +0.79% to +0.61%), but none catches up with dropping.
- **The saved model will have no imputer:** it can only predict complete rows. The test file is complete, so this does not affect F; a new row with a missing value would need an imputer.

SGD

`SGDRegressor` on the features at the carried degree, starting from scikit-learn's defaults with every setting written out: `penalty=None` (the default is a light L2 penalty), the learning rate schedule, `eta0`, `power_t`, `max_iter` and `tol`. `max_iter` is only a ceiling, raised above scikit-learn's default so that only `tol` ends a fit: once its training loss improves by less than `tol` for 5 epochs in a row.

In [9]:
```
SGD_SETTINGS = dict(penalty=None, learning_rate='invscaling', eta0=0.01,
power_t=POWER_T, max_iter=MAX_ITER, tol=TOL, random_state=SEED)  # scikit-learn's
defaults, written out (no penalty, a higher max_iter ceiling)

SETTINGS = {}  # the settings of every SGD run, keyed by name

def sgd(*, name, settings):
        """Cross validate SGD with these settings at the carried degree, imputer and
folds, and keep the settings under name."""
        SETTINGS[name] = settings
        score(name=name, pipeline=build(model=SGDRegressor(**settings), imputer=IMPUTER,
degree=DEGREE), folds=CV_FOLDS)
```

SGD vs Closed Form

In [10]:
```
sgd(name='SGD', settings=SGD_SETTINGS)

table(names=[CLOSED_FORM, 'SGD'])
```

Out[10]:

	validation rmse	train rmse	most epochs
Closed form, degree 2	0.4980	0.4811	<NA>
SGD	0.4987	0.4852	12

Notes / Observations

- **At scikit-learn's defaults SGD lands just above the closed form:** 0.4987 against 0.4980, after at most 12 epochs.

Learning Rate (eta0)

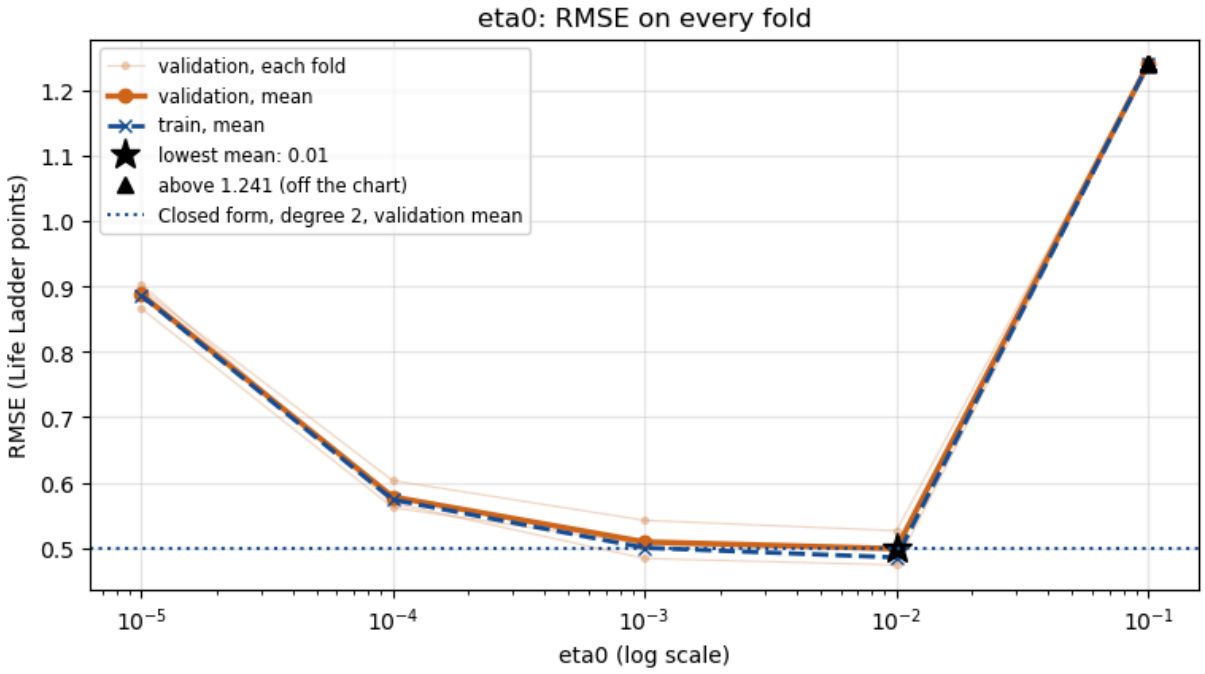
`eta0` is the starting step: under `invscaling` the step shrinks from it as training goes on. It is tried at 5 values, with the schedule at its default, and the default 0.01 is the baseline (rule in Method).

In [11]:
```
names = {eta0: f'eta0={eta0}' for eta0 in ETA0S}
for eta0, name in names.items():
        sgd(name=name, settings={**SGD_SETTINGS, 'eta0': eta0})
```

```
baseline = names[SGD_SETTINGS['eta0']]  # scikit-learn's default, 0.01

display(table(names=[CLOSED_FORM, *names.values()], baseline=baseline))
PLOTTING.parameter_search(
        runs={eta0: RUNS[name] for eta0, name in names.items()},
        parameter='eta0',
        log_x=True,
        reference=(CLOSED_FORM, RUNS[CLOSED_FORM]),
        ymax=YMAX,
        unit=Q1_CONSTS.TARGET_UNIT,
);
SGD_SETTINGS = {**SGD_SETTINGS, 'eta0': SETTINGS[carried(baseline=baseline,
names=names.values())]['eta0']}
```

	validation rmse	train rmse	most epochs	% vs baseline
Closed form, degree 2	0.4980	0.4811	<NA>	-0.14%
eta0=1e-05	0.8886	0.8870	95	+78.18%
eta0=0.0001	0.5780	0.5737	48	+15.90%
eta0=0.001	0.5091	0.5000	14	+2.08%
eta0=0.01	0.4987	0.4852	12	baseline
eta0=0.1	diverged	diverged	6	diverged



Notes / Observations

- **The default 0.01 is kept:** every other value is worse, and the closest, 0.001, is 2.08% higher.
- **Only a narrow range works:** 0.1 diverges, and 0.0001 and below stop well short (0.5780 at 0.0001).

Learning Rate Schedule

How the step changes from `eta0` as training goes on, at the carried `eta0`; `invscaling`, the default, is the baseline (rule in Method).

- `constant`: the step stays `eta0`.

- `optimal`: sets its own step from `alpha` and ignores `eta0`.
- `invscaling`: the step is `eta0 / t ** power_t`, shrinking as training goes on.
- `adaptive`: keeps `eta0` and divides it by 5 each time the training loss stalls for 5 epochs.

In [12]:
```
names = {schedule: f'learning_rate={schedule}' for schedule in LEARNING_RATE_SCHEDULES}
for schedule, name in names.items():
        sgd(name=name, settings={**SGD_SETTINGS, 'learning_rate': schedule})
baseline = names[SGD_SETTINGS['learning_rate']]  # scikit-learn's default, invscaling

display(table(names=[CLOSED_FORM, *names.values()], baseline=baseline))
SGD_SETTINGS = {**SGD_SETTINGS, 'learning_rate': SETTINGS[carried(baseline=baseline,
names=names.values())]['learning_rate']}
display(pd.Series(SGD_SETTINGS, name='carried').astype(str).to_frame())
```

	validation rmse	train rmse	most epochs	% vs baseline
Closed form, degree 2	0.4980	0.4811	<NA>	-0.14%
learning_rate=constant	diverged	diverged	6	diverged
learning_rate=optimal	diverged	diverged	51	diverged
learning_rate=invscaling	0.4987	0.4852	12	baseline
learning_rate=adaptive	diverged	diverged	36	diverged

	carried
penalty	None
learning_rate	invscaling
eta0	0.01
power_t	0.25
max_iter	10000
tol	0.001
random_state	42

Notes / Observations

- **`invscaling`, the default, is the only schedule that does not diverge:** it shrinks the step from the first update. `constant` and `adaptive` start at a step of 0.01 and do not shrink it early, and both diverge; `optimal`, which sets its own step and ignores `eta0`, diverges too.

Regularization

The carried SGD with a penalty: Ridge (`l2`), Lasso (`l1`) and Elastic Net (`elasticnet`, with `l1_ratio` fixed at 0.5, an even mix of both). `alpha`, the penalty's strength, is tried at 3 values a hundred times apart. Every run is compared with the same SGD without a penalty.

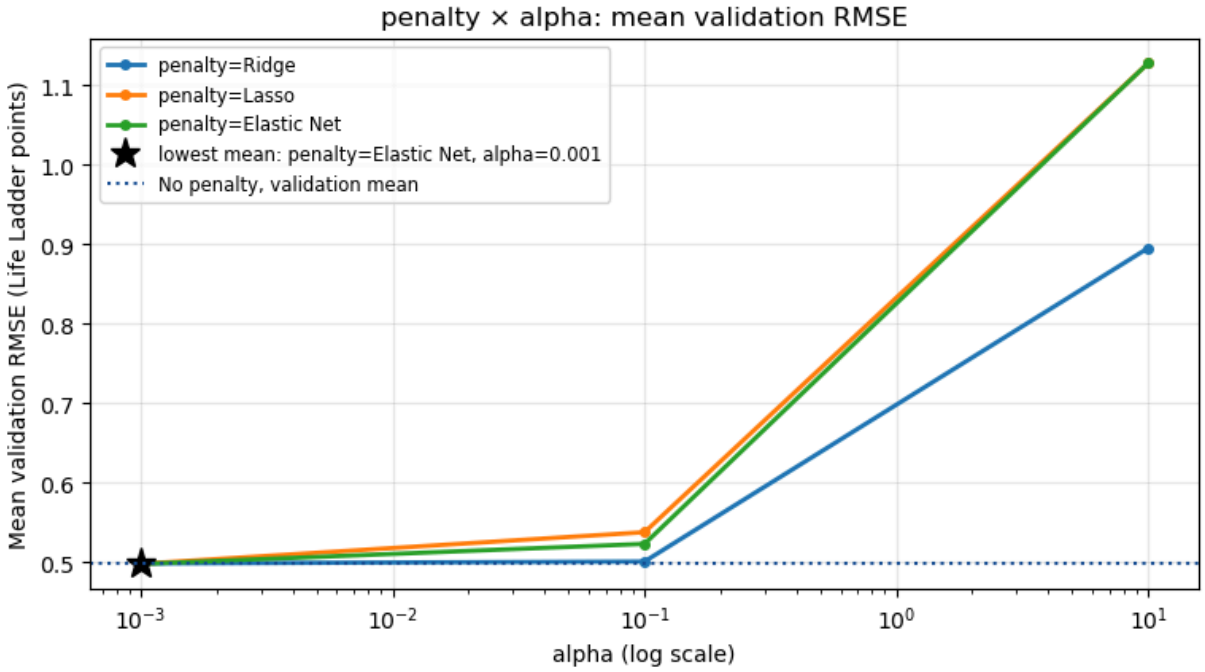
In [13]:
```
PENALTIES = {'Ridge': 'l2', 'Lasso': 'l1', 'Elastic Net': 'elasticnet'}
sgd(name='No penalty', settings=SGD_SETTINGS)

REGULARIZED = {(label, alpha): f'{label}, alpha={alpha}' for label in PENALTIES for
alpha in ALPHAS}
for (label, alpha), name in REGULARIZED.items():
        sgd(name=name, settings={**SGD_SETTINGS, 'penalty': PENALTIES[label], 'alpha':
alpha, **({'l1_ratio': L1_RATIO} if label == 'Elastic Net' else {})})

display(table(names=['No penalty', *REGULARIZED.values()]))
```

```
PLOTTING.parameter_cross(
        runs={pair: RUNS[name] for pair, name in REGULARIZED.items()},
        lines='penalty',
        x='alpha',
        log_x=True,
        reference=('No penalty', RUNS['No penalty']),
        ymax=YMAX,
        unit=Q1_CONSTS.TARGET_UNIT,
);
```

	validation rmse	train rmse	most epochs
No penalty	0.4987	0.4852	12
Ridge, alpha=0.001	0.4987	0.4852	12
Ridge, alpha=0.1	0.5011	0.4904	12
Ridge, alpha=10	0.8955	0.8958	12
Lasso, alpha=0.001	0.4982	0.4853	12
Lasso, alpha=0.1	0.5380	0.5352	8
Lasso, alpha=10	1.1280	1.1279	6
Elastic Net, alpha=0.001	0.4981	0.4848	12
Elastic Net, alpha=0.1	0.5233	0.5188	8
Elastic Net, alpha=10	1.1280	1.1279	6



Notes / Observations

- **A weak penalty changes nothing:** at alpha 0.001 every penalty is within 0.0006 of no penalty (0.4987).
- **L1 bites first:** at alpha 0.1 Lasso rises to 0.5380 while Ridge stays near no penalty (0.5011); Elastic Net lands between them.
- **A strong penalty underfits:** at alpha 10 Ridge's training and validation RMSE rise together (0.8958 and 0.8955), and Lasso and Elastic Net predict the mean (1.1280).

Selected Model

The homework asks for one regularized model, so the 9 regularized runs are ranked by mean validation RMSE and the lowest is selected.

When none of the penalties improves on the model without one, the smallest `alpha` tends to win, because it changes the model the least. The selected model is then close to the model without a penalty: it is picked because the homework requires a regularized one, not because the penalty helps.

In [14]:
```
SELECTED = lowest(names=REGULARIZED.values())
display(pd.Series({**SETTINGS[SELECTED], 'degree': DEGREE},
name=SELECTED).astype(str).to_frame())

table(names=['No penalty', SELECTED])
```

	Elastic Net, alpha=0.001
penalty	elasticnet
learning_rate	invscaling
eta0	0.01
power_t	0.25
max_iter	10000
tol	0.001
random_state	42
alpha	0.001
l1_ratio	0.5
degree	2

Out[14]:

	validation rmse	train rmse	most epochs
No penalty	0.4987	0.4852	12
Elastic Net, alpha=0.001	0.4981	0.4848	12

Notes / Observations

- **Elastic Net at alpha 0.001 is picked as required:** 0.4981 against 0.4987 without a penalty, the smallest alpha, as explained above.

Train on the Entire Training Set

The selected model refit once, with no folds, on the 1,374 complete rows (dropping the incomplete rows is kept, Missing Values) and saved. The saved pipeline takes the 8 raw attributes and builds its own columns, at the carried degree. It has no imputer, so it can only predict complete rows.

In [15]:
```
X_kept, y_kept = (X.iloc[complete], y.iloc[complete]) if IMPUTER is None else (X, y)  #
dropping won: the complete rows only
final = build(model=SGDRegressor(**SETTINGS[SELECTED]), imputer=IMPUTER,
degree=DEGREE).fit(X_kept, y_kept)
Q1_CONSTS.MODEL_OUTPUT_DIR.mkdir(exist_ok=True)
joblib.dump(final, Q1_CONSTS.MODEL_OUTPUT_DIR / 'polynomial_regression.joblib')
```

```
pd.Series({
        'training rows': len(X_kept),
        'epochs run': PIPELINES.final_model(pipeline=final).n_iter_,
        'train rmse': mean_squared_error(y_kept, final.predict(X_kept), squared=False),
}, dtype=object).to_frame(SELECTED)  # object: keeps the counts whole numbers
```

Out[15]:

	Elastic Net, alpha=0.001
training rows	1374
epochs run	10
train rmse	0.4855

Notes / Observations

- **The full fit matches the folds:** on the 1,374 complete rows its training RMSE is 0.4855, against 0.4848 averaged over the folds.

Training and Validation Loss by Epoch

What the chart is for: following training and validation RMSE epoch by epoch, to see when training settles and whether more epochs start to overfit (validation RMSE turning up while training RMSE keeps falling). Each chart shows the selected model on every fold, for as many epochs as it ran on its longest fold.

Two ways to draw it:

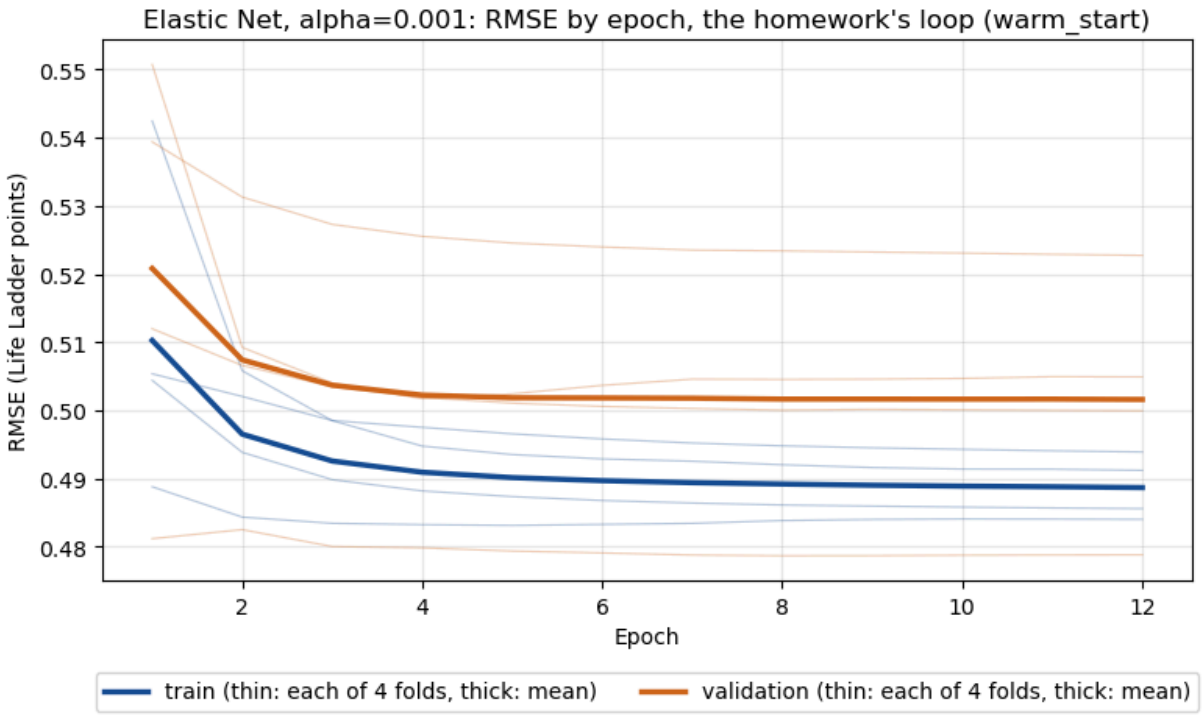
- **The homework's loop:** a copy of the selected model with `max_iter=1` and `warm_start=True`, fit once per epoch on each fold's training rows, recording training and validation RMSE after each `fit` call.
- **The exact curve:** epoch k is a fresh fit of the selected model with `max_iter=k` and the same seed, which is exactly the first k epochs of its real training. A fold's line goes flat once `tol` stops it.

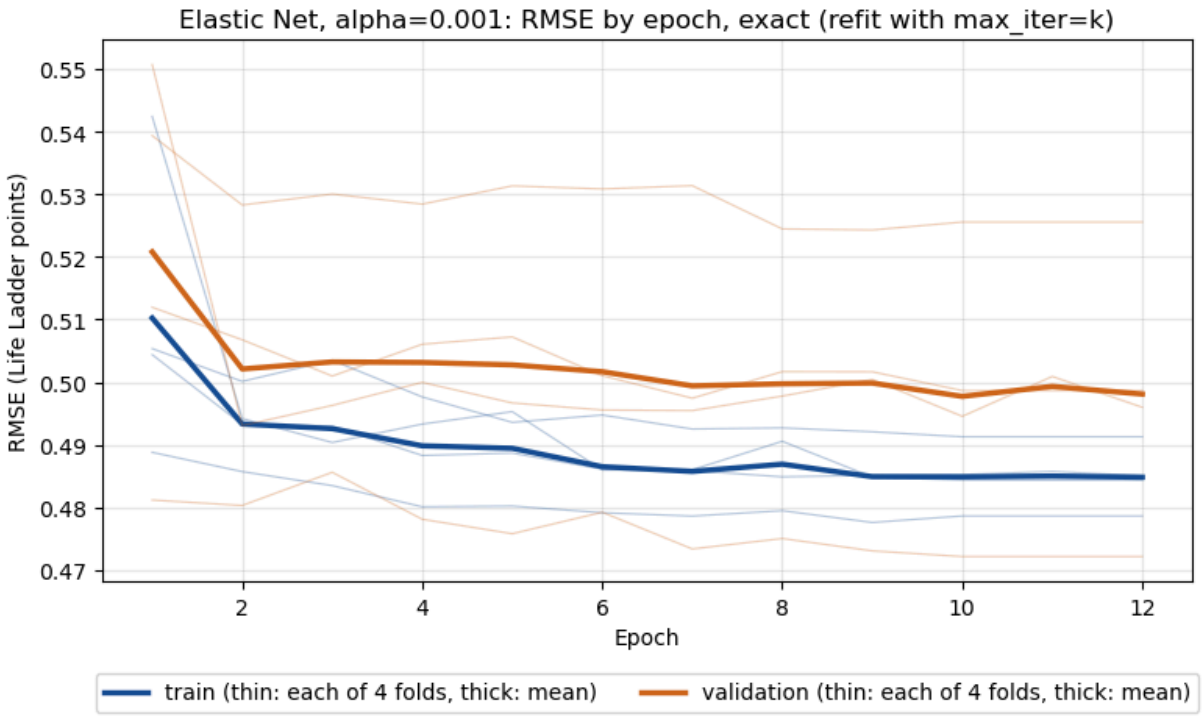
Why they differ: `warm_start=True` carries only the coefficients from one `fit` call to the next. Everything else starts over:

	One long fit (how the selected model trained)	The homework's loop (one `fit` call per epoch)
Coefficients	carry on	carry on (`warm_start=True`)
Step under `constant`	`eta0`	`eta0`, the same
Step under `invscaling` (`eta0 / t ** power_t`, `t` counts the rows seen)	keeps shrinking	starts again at `eta0` every epoch: `fit` sets `t` back to 1
Step under `adaptive` (divided by 5 after 5 epochs without improvement)	shrinks after each stall	stays at `eta0`: a 1-epoch fit can never stall 5 times
Order the rows are visited in	a new shuffle every epoch	the same shuffle every epoch: each `fit` starts from the seed

So the loop replays the real training only under `constant`, and even then it visits the rows in one fixed order. Under the other schedules it trains a different model: one whose step never gets smaller than where a single epoch leaves it. The exact curve shows how the selected model trained; the loop is drawn because the homework asks for it.

In [16]:
```
EPOCHS = int(RUNS[SELECTED]['epochs'].max())  # as many epochs as the selected model ran
on its longest fold
pipeline = build(model=SGDRegressor(**SETTINGS[SELECTED]), imputer=IMPUTER,
degree=DEGREE)
CURVES = {
        method: PIPELINES.epoch_rmse(pipeline=pipeline, X=X, y=y, folds=CV_FOLDS,
epochs=EPOCHS, method=method)
        for method in ('warm_start', 'refit')
}
for method, label in (('warm_start', "the homework's loop (warm_start)"), ('refit',
'exact (refit with max_iter=k)')):
        PLOTTING.loss_curve(curve=CURVES[method], title=f'{SELECTED}: RMSE by epoch,
{label}', ymax=YMAX, unit=Q1_CONSTS.TARGET_UNIT);
```





Notes / Observations

- **Most of the learning happens in the first 2 epochs:** on the exact curve validation RMSE drops from about 0.521 to about 0.502, then edges down to 0.4981.
- **More epochs do not overfit:** validation RMSE edges down with training RMSE and shows no lasting rise.

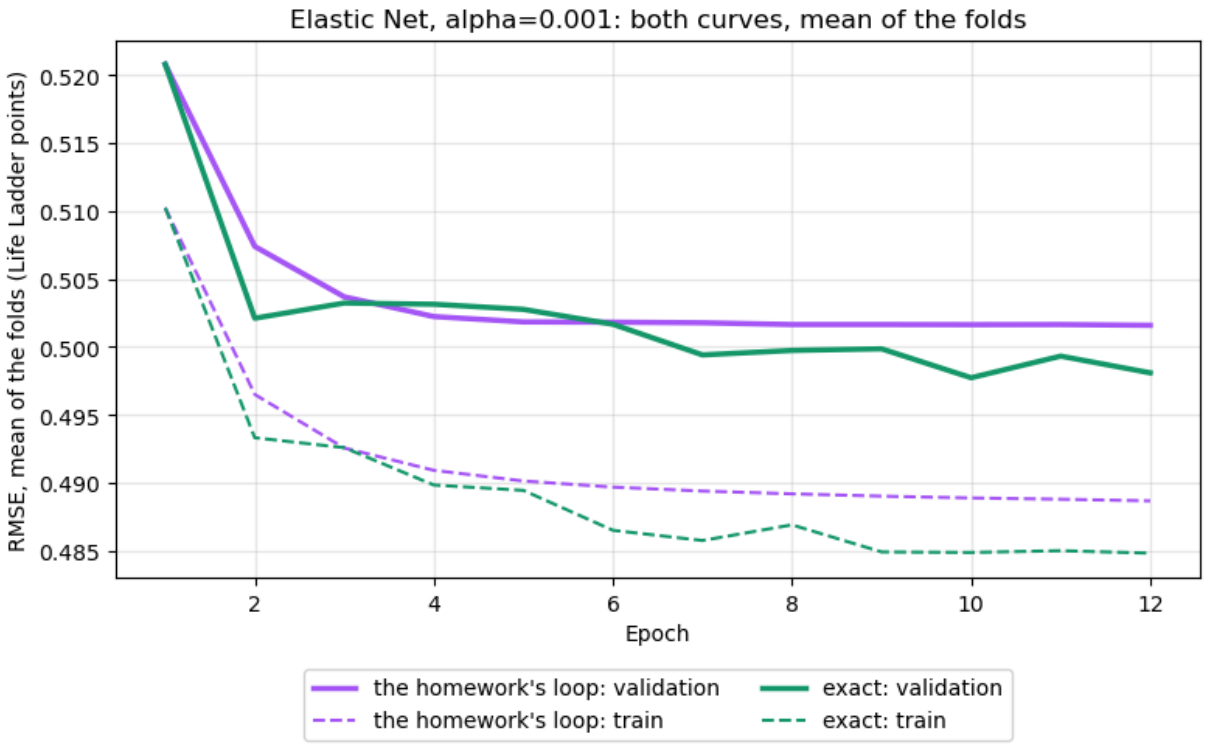
The Two Curves Side by Side

The mean of the folds for both curves on one chart, and the scores each ends on. The exact curve's last epoch is each fold's full fit, so it ends on the selected model's cross validation score from the Selected Model table; the homework's loop does not.

In [17]:
```
PLOTTING.loss_curve_comparison(curves=CURVES, title=f'{SELECTED}: both curves, mean of the folds', ymax=YMAX, unit=Q1_CONSTS.TARGET_UNIT);
pd.DataFrame({
        'cross validation (Selected Model)': RUNS[SELECTED][['validation rmse', 'train rmse']].mean(),
        "homework's loop, last epoch": CURVES['warm_start'].xs(EPOCHS, level='epoch').mean(),
        'exact curve, last epoch': CURVES['refit'].xs(EPOCHS, level='epoch').mean(),
}).T[['validation rmse', 'train rmse']]
```

Out[17]:

	validation rmse	train rmse
cross validation (Selected Model)	0.4981	0.4848
homework's loop, last epoch	0.5016	0.4887
exact curve, last epoch	0.4981	0.4848



Notes / Observations

- **The two curves start together and part at epoch 2:** epoch 1 is the same single pass in both. From epoch 2 the loop restarts `invscaling`'s step at 0.01 every epoch, while the long fit's step keeps shrinking.
- **The loop flattens early and ends higher:** its validation RMSE stops improving around epoch 5 and ends at 0.5016, against 0.4981 for the exact curve. Every epoch starts again with the large step, so the loop gets no closer to the best fit than one epoch can take it.
- **Only the exact curve ends on the cross validation score (0.4981),** so it is the one that shows how the selected model trained.

Overfitting and Underfitting

Validation loss read three ways: by degree (Degree), by penalty (Regularization) and by epoch (Training and Validation Loss by Epoch).

Notes / Observations

- **By degree:** degree 1 underfits, degree 2 fits best, and from degree 3 the closed form overfits (training RMSE falls to 0.4083 while validation RMSE rises to 0.5942).
- **By penalty:** none of the penalties lowers validation RMSE by more than 0.0006, and the larger ones raise training and validation RMSE together, so there is no overfitting for a penalty to cure.
- **By epoch:** the selected model's validation RMSE falls and then holds, so more epochs do not overfit.
- **The selected degree 2 model fits well:** its gap between training and validation RMSE is small (0.4848 against 0.4981).